

TrES-5b

R-0662

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_260529 · created 2026-07-27T17:43:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461189.9192 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.132 +/- 0.036
Transit depth (Rp/Rs)^2	1.73 +/- 0.96 [%]
Orbital Inclination (inc)	84.99 +/- 2.39
Airmass coefficient 1 (a1)	1.202 +/- 0.025
Airmass coefficient 2 (a2)	-0.153 +/- 0.066
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	#2 - [281, 372]
Optimal Method	PSF photometry
Transit Duration (day)	0.072 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.132 ± 0.036 vs 0.142 (deviation 0.28σ)
Duration fit vs published	0.072 d vs 0.0758 d (deviation 0.26σ)
Mid-time O-C	-5.9 min
Depth SNR	3.7
Residual scatter	2.09 %

Light curve

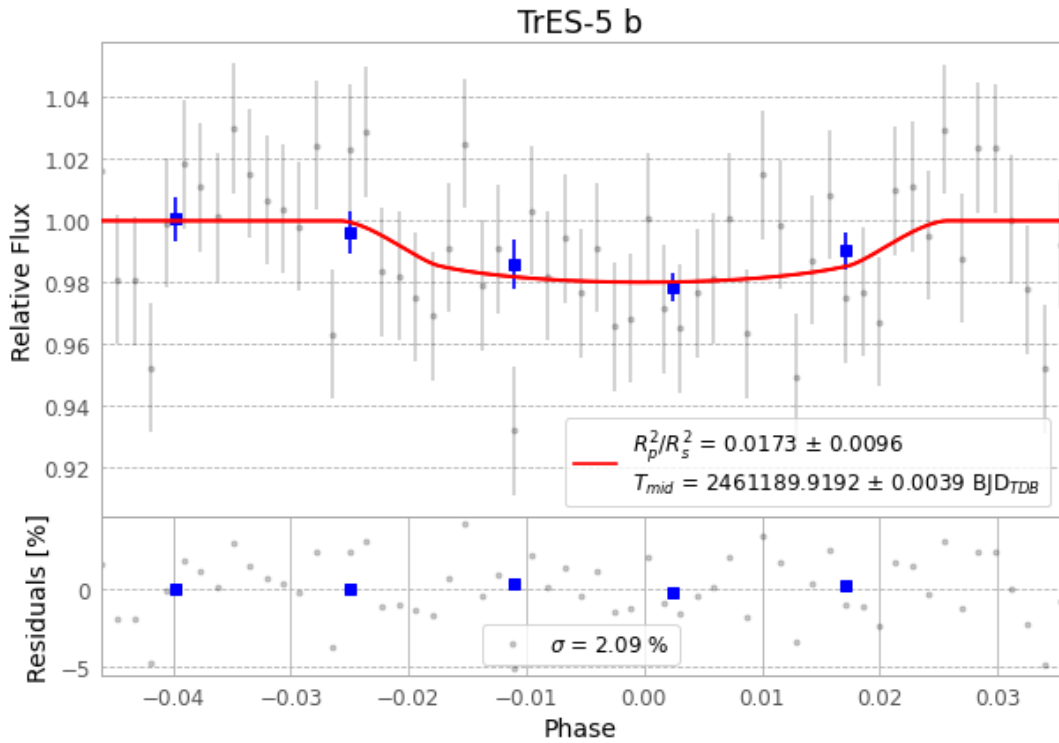


Figure 1 — FinalLightCurve_TrES-5 b_2026-05-29.png (SHA-256 2bae6d7dd3be0438...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee64a80cd748dcec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

59 FITS frames (39 MB), TRES-5260529082415.FITS → TRES-5260529111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-260529.log (ad19d323a10b...), FinalLightCurve_TrES-5 b_2026-05-29.png (2bae6d7dd3be...), FinalLightCurve_TrES-5 b_2026-05-29.csv (bbb1c96f6130...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 17:43Z.